# Stanislav Malashkeevich

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## Research Experience

**Drug Delivery Laboratory, Department of Chemical Enzymology, Lomonosov Moscow State University** *September 2022 – May 2025*

*Research Intern*

- Worked with polymers and their modifications to build micelles capable of carrying and releasing drug compounds.
- Developed a coumarin-based indicator for assaying L-asparaginase activity.
- Applied machine learning to LogP prediction for drug-like molecules, improving $R^2$ by 20% over baseline models.

Co-author of Thermoreversible Gels Based on Chitosan Copolymers as "Intelligent" Drug Delivery System with Prolonged Action for Intramuscular Injection, MDPI Pharmaceutics, 2023 — a study of thermogels for prolonged drug delivery, in particular chitosan-based thermogels.

**Laboratory of Theoretical Chemistry, N. D. Zelinsky Institute of Organic Chemistry, RAS** *December 2024 – Present*

*Research Intern*

- Build and evaluate neural networks for UV–Vis spectrum generation; explored MPNN, Transformer and LLM architectures. Reduced RMSE by 30% and obtained spectra with physically plausible shapes.
- Built a benchmark for LLMs based on chemistry olympiad problems.

## Education

**Lomonosov Moscow State University, Faculty of Chemistry** *2022 – 2028*

*Specialized track "Chemistry of Living Systems and Nanobiotechnology"* *Moscow*

- **Year 2 research project:** Chemoinformatic interpretation of IR spectra of drug-like molecules in micellar systems.
- **Year 3 research project:** Synthesis of the Asp-AMC substrate for studying asparaginase enzymatic activity.
- **Year 4 research project:** Generation of UV–Vis spectra of organic compounds using machine learning across several neural network architectures.

**Specialized Educational and Scientific Center (SESC), Novosibirsk State University** *2021 – 2022*

*Student* *Novosibirsk*

- Prize winner, regional stage of the Russian National Chemistry Olympiad.
- Winner and prize winner of several officially recognized national school olympiads.

**Specialized Educational and Scientific Center (SESC), Ural Federal University** *2020 – 2021*

*Student* *Yekaterinburg*

- Prize winner, regional stage of the Russian National Chemistry Olympiad.
- Winner and prize winner of several officially recognized national school olympiads.

## Awards and Hackathons

**PROSTO Sber ITMO Hackathon** *March 2026*

*Winner (1st place)* *Saint Petersburg*

- Built a team solution for the case "Multi-agent assistant for synthetic organic chemists for laboratory synthesis planning".
- Implemented the NMR spectrum visualization module and a Safety Check system that automatically detects and flags hazardous reagents and toxic compounds; integrated specialized tools connecting the LLM to chemical databases and visualization libraries.
- Event page: prostospb.team/hackathon-26

**Nornickel AI Science Hack** *July 2026*

*Winner (1st place, research track)* *Online*

- Designed and implemented a knowledge-graph architecture (Graph RAG) for retrieving and linking scientific and technical information.
- The resulting pipeline improved the relevance, precision and factual accuracy of LLM answers in complex specialist domains.
- Event page: nornickel-ai-hackathon.ru

## Skills

Experience across the full research cycle: data collection and preprocessing, model training and validation (GNNs, transformers, LLMs), and interpretation of results. Co-author of a publication in a Q1 journal. Seeking a position in ML for Science / drug discovery.

**Machine learning and deep learning:** PyTorch, Transformers, OpenAI API, scikit-learn, VAE, Graph Neural Networks (MPNN), scikit-optimize, NumPy, Matplotlib, Seaborn.
**Chemoinformatics and ML for Science:** RDKit, datamol, Chemprop, SELFIES / SMILES, analysis and generation of UV–Vis / IR / NMR spectra, LLM benchmark construction.
**Molecular dynamics and 3D visualization:** GROMACS, PyMOL, Avogadro, Chimera, Chemcraft.
**Infrastructure and OS:** Linux (Ubuntu, Arch, Manjaro), Bash, Slurm (job submission on HPC/GPU clusters), Git/GitHub, LaTeX, Typst.